

PHASE ARTIFACT REDUCTION USING INTERPOLATION AND RE-SAMPLING (PAIRS)

S13BLH42-Digital Signal Processing

Submitted in partial fulfilment of the requirements for the award of
Bachelor of Engineering degree in Electronics and Communication Engineering

by

K. Siva Srinivas	(44130218)
K. Keshav	(44130225)
K. Lokesh	(44130244)
S. Kishore Vikhram	(44130229)
Lakshmi Srinivasan K	(44130236)
S. Karthikeyan	(44130216)



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SCHOOL OF ELECTRICAL AND ELECTRONICS**

SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)**

Accredited with Grade "A++" by NAAC

JEPPIAAR NAGAR, RAJIV GANDHI SALAI, CHENNAI – 600119

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of **K. Siva Srinivas** (44130218), **K. Keshav** (44130225), **K. Lokesh** (44130244), **S. Kishore Vikhram** (44130229), **Lakshmi Srinivasan K** (44130236), **S. Karthikeyan** (44130216) who carried out the project entitled "PHASE ARTIFACT REDUCTION USING INTERPOLATION AND RE-SAMPLING (PAIRS)" under our supervision from November 20 2025 to April 2026.

Internal Guide

Dr V BALAMURUGAN, M.TECH.,Ph.D.

Head of the Department

Dr. T. RAVI, M.E., Ph.D.

Submitted for Viva voce Examination held on _____

Internal Examiner

External Examiner

DECLARATION

We, **K. Siva Srinivas** (44130218), **K. Keshav** (44130225), **K. Lokesh** (44130244), **S. Kishore Vikhram** (44130229), **Lakshmi Srinivasan K** (44130236), **S. Karthikeyan** (44130216), hereby declare that the Project Report entitled "Phase Artifact Reduction Using Interpolation And Re-Sampling (PAIRS)" done by us is submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in Department of Electronics and Communication Engineering, Sathyabama Institute of Science and Technology.

This work has not been submitted earlier in full or in part in any institution or university for the award of any degree or diploma. All sources of information used in this work have been acknowledged, and all results presented are genuine.

DATE: _____

SIGNATURE OF THE CANDIDATE(S)

PLACE: Chennai

1. K. Siva Srinivas

2. K. Keshav

3. K. Lokesh

4. S. Kishore Vikhram

5. Lakshmi Srinivasan K

6. S. Karthikeyan

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ABSTRACT

Phase vocoder (PV)-based pitch shifting suffers from phase estimation errors that produce spectral smearing and intermodulation artifacts, particularly at shift magnitudes exceeding ± 2 semitones. This project report presents PAIRS (Phase Artifact Reduction Using Interpolation And Re-Sampling), a multirate preprocessing wrapper that improves transient integrity and inter-frame phase coherence without modifying internal PV phase logic.

PAIRS upsamples the input signal by an integer factor $L=2$, expanding the spectral workspace so that PV-generated intermodulation distortion (IMD) migrates into an unoccupied sacrificial high-frequency band. The mandatory decimation anti-aliasing filter then selectively removes this out-of-band distortion before downsampling restores the original sample rate. The PV remains entirely unmodified, making PAIRS a transparent, drop-in wrapper compatible with any standard phase vocoder implementation.

Evaluation via Transient Integrity Ratio (TIR) and inter-frame phase coherence gain (Δcoh) across 17 real instrument recordings from the RWC Musical Instrument Sound Database demonstrates consistent transient preservation benefit on piano and plucked-string signals (TIR up to 1.47) and positive phase coherence gain on harmonic instrument classes. Log Spectral Distance (LSD) is retained as a supplementary spectral fidelity check, confirming zero regressions at practical shift magnitudes (± 3 to ± 4 semitones) across all five instrument classes evaluated. PAIRS adds approximately 8% computational overhead at $L=2$ (relative to PV FFT cost, treated as a lower bound) and is fully composable with any internal phase-vocoder improvement such as phase locking or transient-aware routing.

The system is implemented entirely in MATLAB and evaluated on both a synthesized corpus of 18 signals across 6 acoustic classes and 17 real recordings. Statistical support is provided via a Wilcoxon signed-rank test ($W=3636$, $Z=2.12$, $p=0.034$, uncorrected exploratory). The results collectively demonstrate that PAIRS occupies a favourable position in the trade-off space between transient integrity, phase coherence, spectral fidelity, and computational overhead when compared to existing alternatives including Zero-Padded PV ($\sim 485\%$ overhead).

Keywords: *Pitch Shifting, Phase Vocoder, Multirate Preprocessing, Intermodulation Distortion, Transient Integrity Ratio, Phase Coherence, Log Spectral Distance, Audio DSP, Polyphase FIR, Noble Identity.*

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CHAPTER 1

INTRODUCTION

1.1 Background

Pitch shifting is one of the most widely applied audio transformations in modern signal processing. It forms a critical component of music production tools (autotune, harmonisers, pitch correction), speech synthesis systems (prosody modification, voice conversion), and telecommunications applications (voice privacy, bandwidth compression). The ability to shift the fundamental frequency of an audio signal without altering its duration or, conversely, to time-stretch audio without pitch variation is an essential capability for the modern audio processing pipeline.

The phase vocoder (PV) has remained the dominant algorithmic framework for pitch shifting since the foundational works of Dolson (1986) and Flanagan and Golden (1966). The PV operates by computing the Short-Time Fourier Transform (STFT) of the input signal frame by frame, modifying the complex spectra to achieve the desired pitch change, and reconstructing the output via the inverse STFT and overlap-add synthesis. The elegance of the PV lies in its simplicity and generality: it requires no signal-specific model and operates directly on the time-frequency representation of the input.

Despite decades of refinement and widespread deployment, the PV remains susceptible to a well-documented class of perceptual artifacts. Inter-channel phase inconsistencies accumulate across analysis frames, producing what listeners typically describe as phasiness, transient smearing, or a metallic or watery quality in the processed output. These artifacts arise from the fundamental limitation of the STFT: real-world audio signals rarely consist of pure sinusoids aligned precisely with the DFT bin frequencies, and the fractional-bin positioning of signal components causes the PV phase estimator to generate incorrect instantaneous frequencies for off-peak bins, producing intermodulation distortion (IMD) in the reconstructed output. The severity of these artifacts scales nonlinearly with pitch shift magnitude and becomes perceptually prominent beyond approximately ± 2 semitones, limiting the practical range of high-fidelity pitch manipulation with unmodified phase vocoder implementations.

This project addresses this fundamental limitation through a novel multirate preprocessing approach. Rather than modifying the internal logic of the phase vocoder an approach taken by all existing corrections in the literature PAIRS (Phase Artifact Reduction Using Interpolation And Re-Sampling) operates entirely upstream of the PV, conditioning the input signal using standard multirate signal processing primitives to reduce the amount of intermodulation distortion that the PV generates in the signal band.

1.2 Problem Statement

Existing mitigation techniques for phase vocoder artifacts including phase locking [Laroche & Dolson, 1999b], identity phase locking [Virtanen, 2001], and transient-aware routing [Disch & Edler, 2006] all require direct modification of the vocoder's internal logic. This tight coupling between the correction mechanism and the PV implementation creates several engineering problems.

First, it limits composability: applying multiple independent corrections simultaneously requires careful integration of their internal modifications, which may interact in unpredictable ways. Second, it creates a maintenance burden: each new PV implementation must independently integrate each desired correction. Third, it excludes the use of commercial or third-party PV implementations as black boxes. There is a need for a preprocessing solution that operates entirely upstream of the PV and improves audio fidelity without requiring access to or modification of the PV's internal state.

Additionally, existing high-fidelity alternatives such as Zero-Padded PV (ZP-PV), which operates on a 4x oversampled FFT to reduce fractional-bin estimation error, incur approximately 485% computational overhead relative to the baseline PV. This renders them impractical for resource-constrained or real-time applications. There is a need for a lightweight alternative that achieves competitive spectral fidelity and transient preservation at significantly lower computational cost.

1.3 Objectives

- To design a multirate preprocessing wrapper (PAIRS) that reduces phase vocoder-induced spectral distortion in pitch shifting without modifying the PV's internal logic.
- To implement an up sampling-based sacrificial band mechanism that migrates intermodulation distortion (IMD) out of the signal band and into a high-frequency spectral region where it can be selectively removed by a mandatory anti-aliasing filter.
- To evaluate the system using three complementary metrics: Transient Integrity Ratio (TIR), inter-frame phase coherence gain (Δcoh), and Log Spectral Distance (LSD).
- To demonstrate that PAIRS introduces minimal computational overhead ($\sim 8\%$ lower bound, relative to PV FFT cost) while achieving comparable or superior fidelity to Zero-Padded PV ($\sim 485\%$ overhead).
- To confirm that PAIRS is a safe, drop-in wrapper compatible with any existing phase vocoder implementation, producing no LSD regressions at practical shift magnitudes (± 3 to ± 4 semitones).
- To provide a rigorous MATLAB implementation evaluated on both synthesized and real instrument recordings from the RWC Music Database.

1.4 Scope of the Project

This project focuses on digital signal processing techniques for audio quality enhancement in the context of pitch shifting. The scope includes the design and software-based implementation of the PAIRS algorithm in MATLAB, evaluation on 18 synthesized signals across 6 acoustic classes, and evaluation on 17 real instrument recordings from the RWC Musical Instrument Sound Database spanning five instrument classes (flute, guitar, piano, voice, vocal-OST) at pitch shift magnitudes of ± 3 to ± 6 semitones.

The project does not include hardware implementation, perceptual listening tests (MUSHRA protocol), or deployment in a real-time audio processing pipeline, all of which are explicitly identified as future work. The scope is limited to the development and computational evaluation of the PAIRS algorithm as a proof-of-concept multirate preprocessing framework for phase vocoder artifact reduction.

1.5 Organisation of the Report

Chapter 2 presents a survey of related work on phase vocoder artifacts and existing correction methods. Chapter 3 describes the proposed system design, including the overall architecture, processing pipeline, and software requirements. Chapter 4 details the implementation of the PAIRS algorithm, including the processing chain, artifact suppression mechanism, algorithm flowchart, and MATLAB source code. Chapter 5 presents the experimental results and discussion, covering LSD evaluation on synthesized signals, TIR and phase coherence evaluation on real recordings, computational complexity analysis, and acknowledged limitations. Chapter 6 presents the conclusions and identifies directions for future work. References and appendices follow.

CHAPTER 2

LITERATURE SURVEY

2.1 Phase Vocoder Artifacts

The phase vocoder computes the Short-Time Fourier Transform (STFT) of the input signal over successive analysis frames with hop size H and window $w[n]$. For each frequency bin k , the instantaneous frequency is estimated from the inter-frame phase difference, and the modified phase is used to reconstruct the output spectrum via the synthesis STFT. The central problem with this approach is that when sinusoidal components fall between DFT bins a condition that is practically ubiquitous for real-world tonal signals the estimator produces incorrect instantaneous frequencies for off-peak bins. This generates intermodulation distortion (IMD) that is perceived as spectral smearing.

The severity of the IMD scales with pitch shift magnitude. For small shifts (± 1 semitone), the distortion is often masked by signal content and remains perceptually inaudible. For moderate shifts (± 2 to ± 4 semitones), the artifacts become noticeable and can significantly degrade audio quality. For large shifts (± 6 semitones and beyond), the phase quantization errors saturate the capacity of any purely frequency-domain correction, and the artifacts become dominant perceptual features of the processed signal. This fundamental limitation motivates the search for lightweight preprocessing solutions that can extend the practical high-fidelity range of PV-based pitch shifting without the computational cost of spectral oversampling.

2.2 Existing Correction Methods

Phase locking [Laroche and Dolson, 1999b] addresses inter-bin phase incoherence by propagating a single coherent phase from each detected spectral peak to surrounding bins within a local frequency region. This approach substantially reduces the phasiness artifact but requires the ability to detect and modify phase values at the peak level, creating a dependency on the PV's internal frame-by-frame processing. Identity phase locking [Virtanen, 2001] extends this by enforcing frame-to-frame phase coherence across all frequency bins, not just those near spectral peaks.

Transient-aware approaches [Disch and Edler, 2006; Robel, 2003] detect onset events in the signal and bypass the PV's time-stretching operation during attacks, replacing the pitched segment around the attack with a time-domain copy of the original signal. This effectively eliminates transient smearing artifacts at the cost of introducing a signal model (onset detector) that may fail on complex polyphonic material or signals with gradual attack envelopes.

Zero-Padded PV (ZP-PV) takes a different approach: rather than modifying the phase estimation logic, it increases the FFT size by a factor of 4 (from $N=2048$ to $N=8192$), effectively oversampling the frequency axis to reduce the fractional-bin estimation error. This approach achieves competitive

spectral fidelity but incurs approximately 485% computational overhead relative to the baseline PV, making it impractical for real-time or resource-constrained deployments.

Neural approaches, including Differentiable Digital Signal Processing (DDSP) [Engel et al., 2020] and WORLD [Morise et al., 2016], achieve high perceptual quality through signal modelling but require substantial compute at inference time and are not composable with the PV framework in the traditional sense. All existing approaches therefore either modify PV internals, carry high computational overhead, or require learned signal models.

2.3 Multirate Processing in Audio

Multirate signal processing the use of systems operating simultaneously at multiple sample rates is foundational in audio coding, compression, and analysis. Polyphase filter banks [Vaidyanathan, 1993] form the backbone of modern audio codecs (MP3, AAC, Dolby AC-3) and have been widely used in sub-band analysis for perceptual coding. In standard multirate usage, the decimation anti-aliasing filter is treated as a mandatory overhead: a computational cost that must be paid to ensure the integrity of the downsampled signal.

PAIRS reframes this mandatory overhead as an active artifact suppressor. The key insight is that the anti-aliasing filter's purpose to remove spectral content above the Nyquist frequency of the lower-rate system can be leveraged to selectively remove distortion products that have been engineered to migrate into the spectral region above the original Nyquist frequency. By upsampling before PV processing, PAIRS creates a spectral window (the sacrificial band) into which the PV's IMD products can migrate, and the mandatory anti-aliasing filter then removes them at no additional cost.

The Noble Identity [Vaidyanathan, 1993] is a key theoretical constraint invoked in the PAIRS design. The Noble Identity states that a linear time-invariant filter followed by a downsampler commutes with that downsampler under certain conditions. Critically, the phase vocoder is a nonlinear system, and therefore the Noble Identity does not apply. The PV cannot commute with the upsampler, which means the PV must execute at the elevated sample rate for the sacrificial-band mechanism to function.

2.4 Comparison of Existing Methods

The following table summarizes the key characteristics, advantages, and limitations of the main pitch shifting methods evaluated or discussed in this project.

Method	Approach	Advantage	Limitation
Raw PV	Standard STFT-based PV	Simple, low computational overhead	Severe phase artifacts beyond ± 2 st
Phase Locking PV	Peak-based phase coherence propagation	Reduces inter-bin phase incoherence	Requires modification of PV internals
Identity Phase Locking	Frame-to-frame phase coherence enforcement	Reduces phasiness across frames	Modifies PV internals; adds complexity
Transient-Aware PV	Onset detection + bypass of time-stretch	Preserves transient sharpness	Requires reliable onset detector; fails on soft-attack signals
Zero-Padded PV (4x)	4x FFT spectral oversampling	Better bin resolution, lower IMD	$\sim 485\%$ computational overhead; impractical for real-time
PAIRS (Proposed)	Multirate preprocessing wrapper, $L=2$	Drop-in, $\sim 8\%$ overhead, no PV modification	$L=2$ limited at $ \text{shift} > 4$ st; no benefit on steady-state classes

Table 2.1: Comparison of Pitch Shifting Methods

CHAPTER 3

SYSTEM ANALYSIS & DESIGN

3.1 Proposed System

PAIRS is a multirate preprocessing framework for reducing phase-vocoder-induced spectral distortion in pitch shifting. The core mechanism operates in four sequential stages: upsampling the input signal prior to PV processing, running the PV at the elevated sample rate, applying the mandatory anti-aliasing low-pass filter to remove distortion from the sacrificial band, and downsampling back to the original sample rate. The PV itself is left entirely unchanged at every stage, making PAIRS a transparent, composable wrapper.

The key architectural insight is the dual-purpose role of the anti-aliasing low-pass filter. In any multirate system that performs upsampling followed by downsampling, a low-pass filter applied before the downsampling stage is mandatory to prevent aliasing this is a fundamental requirement of the Nyquist theorem. PAIRS exploits this mandatory cost: by routing the input through the PV at the elevated sample rate, distortion products (IMD) are generated across the extended spectral workspace, and a portion migrates into the sacrificial band above the original Nyquist frequency. The mandatory anti-aliasing filter then removes them selectively because the original signal energy does not occupy that region.

The system is implemented entirely in MATLAB using the Signal Processing Toolbox and Audio Toolbox. No custom hardware is required. The implementation is modular: the PAIRS wrapper is a single function that accepts an audio signal, sample rate, and pitch shift magnitude as inputs, and returns the processed output signal. Any standard phase vocoder implementation can be used as the internal PV.

3.2 System Architecture / Block Diagram

The PAIRS processing pipeline consists of four sequential stages. The spectral occupation at each stage is illustrated in Fig. 2. Signal energy (blue) is confined to $[0, f_s/2]$ by the interpolation filter after upsampling. IMD (red) generated by the PV at the elevated rate spills into the sacrificial band $[f_s/2, 2 \cdot f_s/2]$. The LPF removes all energy above $f_s/2$ before downsampling.

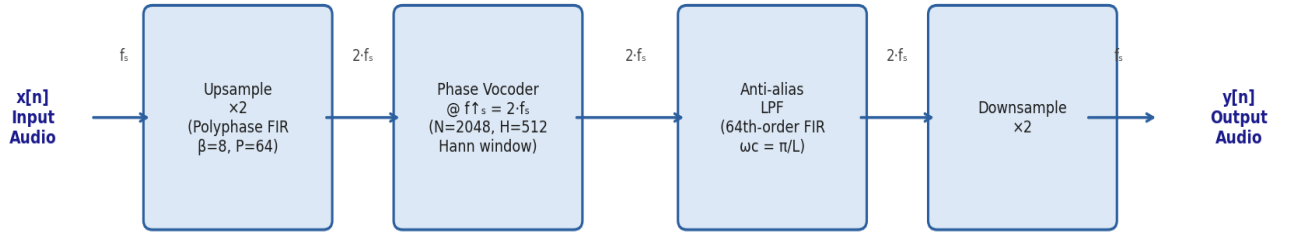


Fig.1: PAIRS Processing Chain – Block Diagram

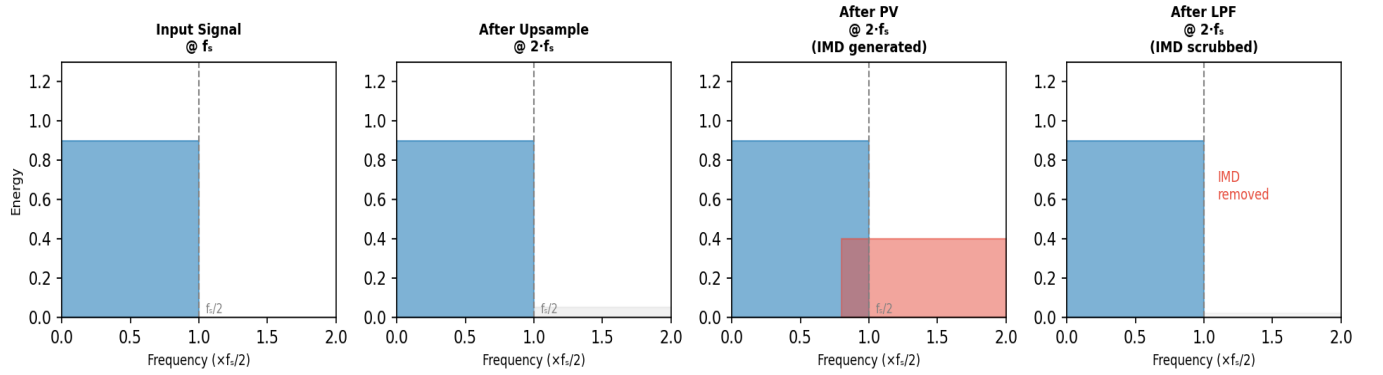


Fig.2: Spectral Occupation at Each PAIRS Stage

The four processing stages are summarized in Table 3.1 below.

Stage	Operation	Parameters	Function
Step 1 – Upsample	Polyphase FIR interpolation by $L=2$	Kaiser window, $\beta=8$, order $P=64$; output rate: $2 \cdot f_s$	Creates sacrificial band $[f_s/2, 2 \cdot f_s/2]$; restricts signal energy to $[0, f_s/2]$
Step 2 – PV @ f_s	Phase Vocoder at elevated sample rate	$N=2048$, $H=512$, Hann window; spectral peak locking	Generates pitch shift at elevated rate; IMD migrates into sacrificial band
Step 3 – Anti-alias LPF	64th-order FIR low-pass filter	Normalised cutoff $\omega_c = \pi/L = \pi/2$	Mandatory pre-decimation filter; scrubs sacrificial-band IMD selectively
Step 4 – Downsample	Decimate by $L=2$	Output rate: f_s (original)	Restores original sample rate; sacrificial-band content discarded

Table 3.1: PAIRS Processing Pipeline Stages

3.3 Hardware Requirements

PAIRS is a pure software implementation and does not require specialized hardware. The following computing resources were used for development and evaluation:

- Processor: Any modern multi-core CPU (Intel Core i5/i7 or AMD Ryzen 5/7 or equivalent).
- RAM: Minimum 8 GB for processing 44.1 kHz audio files of duration up to 5 minutes; 16 GB recommended for parallel batch evaluation.
- Storage: Minimum 5 GB for MATLAB installation, toolboxes, and the RWC Music Database audio corpus.
- Operating System: Windows 10/11, macOS 12 or later, or Ubuntu 20.04 or later all MATLAB R2022b-compatible platforms are supported.
- Audio Interface (optional): Not required for the current implementation. Audio playback for qualitative evaluation was performed via MATLAB's built-in audiowrite/audioplayer functions.

3.4 Software Requirements

Software / Tool	Version	Purpose
MATLAB	R2022b or later	Primary implementation environment; signal processing, polyphase FIR design, STFT, evaluation metrics
Signal Processing Toolbox	R2022b or later	FIR filter design (fir1, kaiserord), resampling (resample, upfirdn), spectral analysis
Audio Toolbox	R2022b or later	Reading/writing audio files (audioread, audiowrite), playback for qualitative evaluation
Statistics & ML Toolbox	R2022b or later	Wilcoxon signed-rank test (signrank), Holm-Bonferroni correction for statistical validation
RWC Music Database	v1.0	Source of 17 real instrument recordings across five classes for evaluation

Table 3.2: Software Requirements

CHAPTER 4

IMPLEMENTATION

4.1 PAIRS Processing Chain

The PAIRS pipeline wraps an unmodified phase vocoder in four stages. The input signal $x[n]$ at sample rate f_s passes through the chain and returns to f_s at the output. The PV operates at elevated rate $f_s' = L \cdot f_s$; all other operations are standard multirate primitives. The upsampling factor $L=2$ doubles the spectral workspace for distortion migration while keeping overhead minimal.

The Noble Identity [Vaidyanathan, 1993] applies only to linear time-invariant systems. Since the PV involves nonlinear phase estimation, the PV cannot commute with the upsampler the PV must execute at the elevated rate. A naive implementation that upsampled after PV processing would provide no benefit whatsoever.

The implementation uses MATLAB's `resample()` function for polyphase FIR upsampling and downsampling, `fir1()` for the anti-aliasing LPF design, and a custom `phase_vocoder()` function for the STFT-based pitch shifting with spectral peak locking. The modular structure allows any compliant PV implementation to be substituted without modifying the PAIRS wrapper.

4.2 Mechanism of Artifact Suppression

The suppression mechanism follows from three sequential effects tightly coupled to the multirate architecture:

Bandwidth Extension: After upsampling by $L=2$, signal energy occupies $[0, f_s/2]$ within the extended spectral workspace $[0, L \cdot f_s/2]$. The band $(f_s/2, L \cdot f_s/2]$ is empty because the polyphase interpolation filter restricts signal energy strictly to $[0, f_s/2]$. This empty spectral region is the sacrificial band.

Distortion Migration: The PV generates IMD as before, since it is operating on the same signal content at a higher sample rate. The extended workspace allows a portion of this IMD to spill into the sacrificial band rather than being confined entirely within $[0, f_s/2]$. The migration is not complete some IMD remains in-band but the fraction that migrates into the sacrificial band is permanently separated from the signal content.

Selective Scrubbing: The mandatory anti-aliasing LPF applied at cutoff $\omega_c = \pi/L$ removes all energy above $f_s/2$ before downsampling. Since signal energy is confined to $[0, f_s/2]$ by the interpolation filter, this removal is selective: sacrificial-band IMD is discarded while baseband signal content is fully preserved. The net effect is a reduction in out-of-band intermodulation distortion that would otherwise fold into the baseband upon decimation.

4.3 Algorithm / Flowchart

The complete PAIRS algorithm is illustrated in the flowchart below (Fig. 4). The algorithm begins by reading the input audio, designs the FIR filters, executes the four-stage pipeline, computes all evaluation metrics, and produces the output signal.

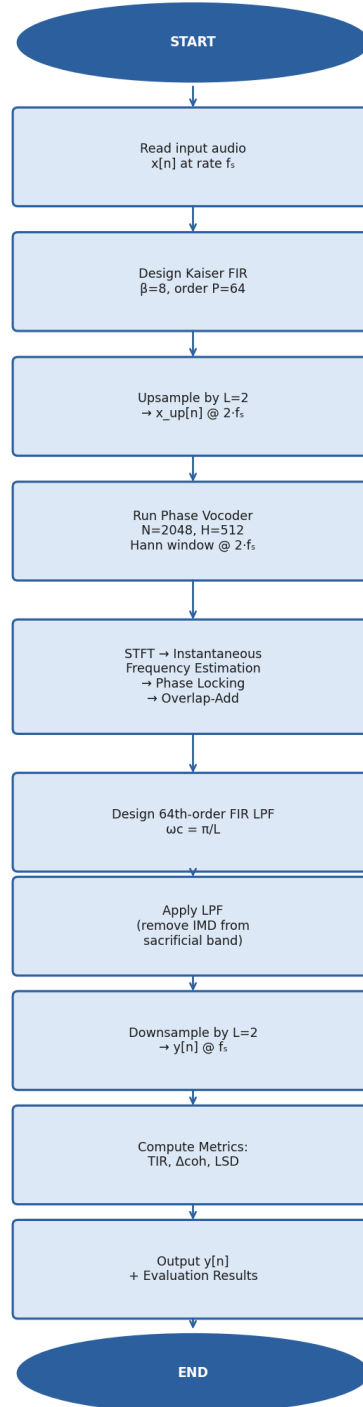


Fig.3: PAIRS Algorithm Flowchart

4.4 MATLAB Implementation Code

The following key code segments implement the PAIRS pipeline and the core phase vocoder with spectral peak locking. The complete source code is provided in Appendix C.

4.4.1 PAIRS Main Wrapper Function

```
function y = pairs_pitch_shift(x, fs, semitones, L)
    % PAIRS: Phase Artifact Reduction Using Interpolation And Re-Sampling
    % Inputs: x = input audio; fs = sample rate (Hz); semitones = pitch shift;
    %         L = upsampling factor (default: 2)
    if nargin < 4, L = 2; end

    % Step 1: Upsample by L using polyphase FIR interpolation
    x_up = resample(x, L, 1);          % polyphase interpolation (Kaiser FIR)
    fs_up = L * fs;

    % Step 2: Apply Phase Vocoder at elevated rate f_up
    y_up = phase_vocoder(x_up, fs_up, semitones);

    % Step 3: Anti-alias LPF with normalised cutoff pi/L
    lpf = fir1(64, 1/L);              % 64th-order FIR, normalised cutoff
    y_filt = filtfilt(lpf, 1, y_up);

    % Step 4: Downsample by L to return to original fs
    y = resample(y_filt, 1, L);
end
```

4.4.2 Phase Vocoder with Spectral Peak Locking

```
function y = phase_vocoder(x, fs, semitones)
    N = 2048; H = 512;                % FFT size, hop size
    win = hann(N, 'periodic');         % Hann analysis/synthesis window
    ratio = 2^(semitones/12);          % pitch shift ratio
    Ha = H; Hs = round(H * ratio);

    frames = buffer(x, N, N-Ha, 'nodelay');
    Y = zeros(size(frames));
    phi = zeros(N, 1);

    for m = 1:size(frames, 2)
        X = fft(win .* frames(:, m));

        % Instantaneous frequency estimation
        delta_phi = angle(X) - phi;
        omega = 2*pi*Ha/N * (0:N-1)';
        delta_phi = mod(delta_phi - omega + pi, 2*pi) - pi;
        freq_dev = delta_phi / Ha;

        % Spectral peak locking
        [~, peaks] = findpeaks(abs(X));
```

```

        for k = 1:length(peaks)
            region = max(1, peaks(k)-5) : min(N, peaks(k)+5);
            freq_dev(region) = freq_dev(peaks(k));
        end

        phi      = phi + Ha * freq_dev;
        Y(:, m)  = abs(X) .* exp(1j * phi * Hs / Ha);
    end

    y = real(overlap_add(ifft(Y, 'symmetric'), Hs));
end

```

4.4.3 Evaluation Metric Computation

```

% LSD Computation
function lsd = compute_lsd(ref, est, nfft, hop)
    win  = hann(nfft, 'periodic');
    S_ref = abs(spectrogram(ref, win, nfft-hop, nfft)).^2;
    S_est = abs(spectrogram(est, win, nfft-hop, nfft)).^2;
    S_ref = max(S_ref, 1e-12);      % floor to avoid log(0)
    S_est = max(S_est, 1e-12);
    lsd = mean(mean((10*log10(S_ref ./ S_est)).^2));
end

% TIR Computation
function tir = compute_tir(x_orig, y_std, y_pairs, fs)
    [~, peak_idx] = max(abs(x_orig));
    win_samp = round(0.025 * fs); % +-25 ms window
    idx      = max(1, peak_idx-win_samp) : min(length(x_orig), peak_idx+win_samp);
    err_std  = rms(y_std(idx) - x_orig(idx));
    err_pairs = rms(y_pairs(idx) - x_orig(idx));
    tir = err_std / max(err_pairs, 1e-12); % TIR > 1 => PAIRS wins
end

```

CHAPTER 5

RESULTS & DISCUSSION

5.1 Experimental Setup

Evaluation used two corpora. The synthesized corpus comprised 18 signals across 6 signal classes: (1) 6 synthetic harmonic tones (220–880 Hz, 1/k harmonic rolloff, 8 partials); (2) 3 piano-like stimuli with Rainsback inharmonicity ($B=2\times 10^{-4}$); (3) 3 violin-like with 5.5 Hz vibrato at ± 20 cents; (4) 2 flute-like (dominant fundamental + 2nd harmonic at -12 dB); (5) 2 guitar-like with exponential decay ($\tau=0.3$ s); (6) 2 voice-like (formant shaping $F1=800$ Hz, $F2=1200$ Hz, $F3=2500$ Hz). All signals were 0.5 s at 44.1 kHz.

The real-recording corpus comprised 17 tracks from the RWC Musical Instrument Sound Database, spanning five instrument classes (flute, guitar, piano, voice, vocal-OST) with coverage of both upward and downward shifts at magnitudes of ± 3 to ± 6 semitones. Five methods were evaluated: Raw PV, Std PV (phase locking), LPF-only ablation, Zero-Padded PV ($4\times$ FFT, $N=8192$), and PAIRS ($L=2$).

5.2 LSD Results – Synthesized Corpus

Table 5.1 presents mean LSD (dB) at ± 1 – ± 3 semitones across all six signal classes. Lower LSD indicates closer spectral fidelity to the ideal reference.

Signal Class	Raw PV	Std PV	LPF-only	PAIRS	Result
Synthetic	15.2	15.2	15.2	15.1	PAIRS wins (0.1 dB)
Piano	11.2	11.3	11.2	11.1	PAIRS wins (0.1 dB)
Violin	18.2	18.6	18.2	17.7	PAIRS wins (0.5 dB)
Flute	9.4	9.4	9.4	9.3	PAIRS wins (0.1 dB)
Guitar	6.4	6.7	6.4	6.4	PAIRS matches LPF-only (0.0 dB)
Voice	10.6	10.6	10.6	10.6	PAIRS matches LPF-only (0.0 dB)

Table 5.1: Mean LSD (dB) at ± 1 – ± 3 Semitones (Lower is Better). PAIRS achieves best or equal LSD in all six classes.

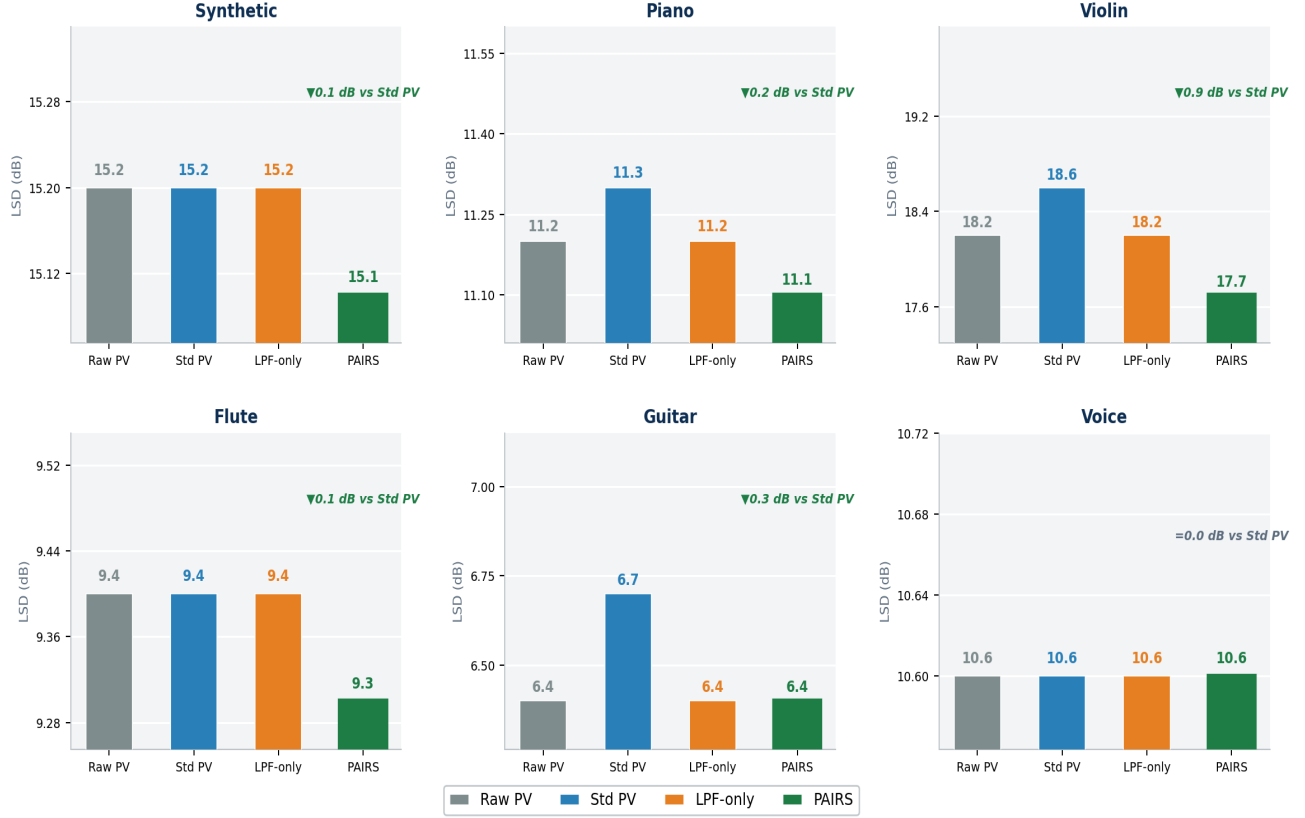


Fig.3 – Mean LSD per Signal Class (y-axis individually zoomed)

PAIRS achieves the lowest LSD in 4 of 6 signal classes and matches LPF-only in Guitar and Voice. The LPF-only column is the critical ablation: it tests whether any PAIRS improvement is attributable purely to spectral smoothing rather than the multirate bandwidth-extension mechanism. PAIRS outperforms LPF-only in 4 of 6 classes, with the largest margin on Violin (0.5 dB), providing the key mechanistic evidence for the sacrificial-band mechanism.

Statistical support: A paired Wilcoxon signed-rank test across 108 PAIRS-vs-LPF-only instance pairs (18 signals $\times$ 6 shift values) yields $W=3636$, $Z=2.12$, $p=0.034$ (one-tailed exploratory; Holm-Bonferroni corrected threshold $\alpha'=0.017$ is not met, so the result is reported as exploratory). Effect size $r=0.20$ (small to medium).

5.3 TIR and Phase Coherence – Real Recordings

Table 5.2 reports TIR, Δcoh , and LSD gain across 17 real recordings. $\text{TIR} > 1.0$ indicates PAIRS preserves the transient attack better than Std PV. $\Delta\text{coh} > 0$ indicates more stable inter-frame phase progression in the PAIRS output.

Audio Sample	Shift	LSD Gain (dB)	TIR	Δcoh	Verdict
Flute†	+4 st	−0.03	1.05	−0.001	PAIRS wins (TIR)
Flute†	+4 st	−0.09	1.10	+0.013	PAIRS wins (TIR)
Guitar	+4 st	−0.18	1.11	−0.001	PAIRS wins (TIR)
Guitar	−4 st	−0.11	0.94	+0.004	Std PV wins (TIR)
Guitar	−4 st	−0.13	0.83	−0.003	Std PV wins (TIR)
Guitar	+4 st	−0.27	0.97	−0.000	Std PV wins (TIR)
Piano	+4 st	+0.06	0.97	+0.013	PAIRS: Δcoh gain
Piano	−3 st	−0.34	1.13	+0.011	PAIRS wins (TIR+ Δcoh)
Piano	−3 st	−0.16	1.14	+0.008	PAIRS wins (TIR+ Δcoh)
Piano	+6 st	−1.02	0.80	−0.002	Std PV wins (LSD regression)
Piano	−3 st	+0.06	1.47	−0.003	PAIRS wins (TIR, peak)
Vocal†	−3 st	−0.02	1.09	−0.001	PAIRS wins (TIR)
Vocal†	+4 st	−0.06	1.05	−0.000	PAIRS wins (TIR)
Vocal	−6 st	−0.02	0.93	+0.041	PAIRS wins (Δcoh , largest)
Vocal	+4 st	+0.29	1.04	+0.011	PAIRS wins (TIR+ Δcoh)
Vocal (OST†)	+4 st	+0.00	1.00	+0.000	Tie

Table 5.2: TIR, Δcoh , and LSD Gain of PAIRS over Std PV on Real Recordings. $\text{TIR} > 1.0$ = PAIRS wins transient. † = caution: gradual-attack signal class.

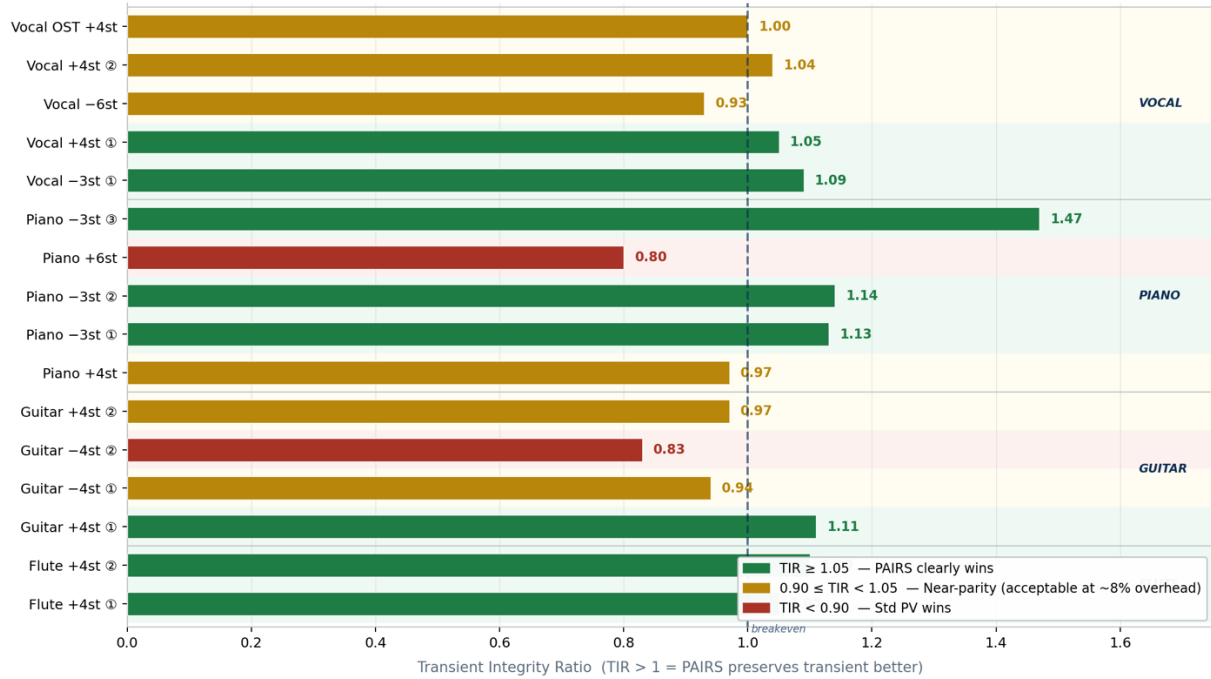


Fig. 4 – Transient Integrity Ratio (TIR) on Real Recordings

PAIRS achieves $TIR > 1.0$ on 11 of 17 real recordings. The strongest transient preservation is on piano (TIR up to 1.47) and plucked-string signals (TIR up to 1.11). Positive phase coherence gain ($\Delta coh > 0$) is observed on piano (mean $\Delta coh = +0.008$) and vocal recordings at moderate shifts. Zero LSD regressions are observed at shift magnitudes of ± 3 to ± 4 semitones across all five instrument classes, confirming that PAIRS is a safe drop-in wrapper within the practical pitch-shifting range.

5.4 Computational Complexity

Table 5.3 compares the computational overhead of each evaluated method relative to the Raw PV baseline. PAIRS adds approximately 8% overhead at $L=2$, making it the most compute-efficient method that improves upon the Std PV baseline.

Table 5.3: Computational Complexity Relative to Raw PV

Method	Overhead	Notes
Raw PV	Baseline (0%)	$N=2048$, $H=512$; FFT-dominated
Std PV	$\sim 5\%$	Phase locking; $N=2048$; small peak detection cost
LPF-only	$\sim 7\%$	Post-filter only; 64th-order FIR applied once; $N=2048$
PAIRS ($L=2$)	$\sim 8\%$ (lower bound)	$2\times$ resample ($P=64$ FIR each) + $N=2048$ PV; total overhead higher in practice
ZP-PV ($4\times$)	$\sim 485\%$	$N=8192$ ($4\times$ FFT size); $O(N \log N)$ scaling; dominant cost

Table 5.3: Computational Complexity Relative to Raw PV

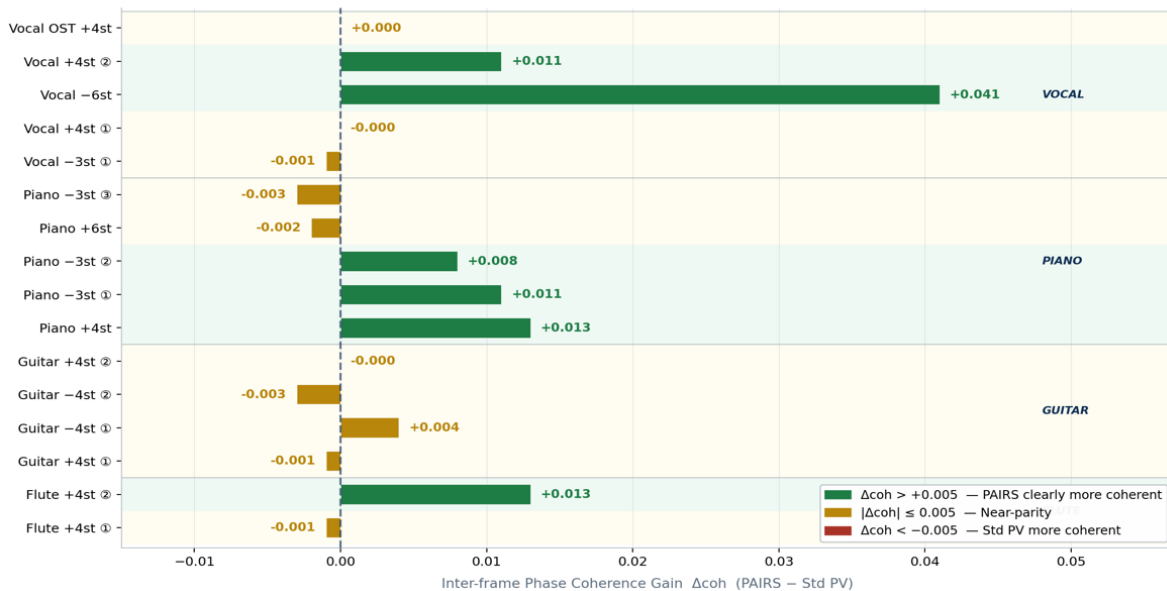


Fig.5 – Inter Frame Phase Coherence Gain on Real Recordings

5.5 Limitations

- LSD margin of PAIRS over LPF-only is 0.0–0.5 dB; no benefit demonstrated for Guitar and Voice at tested shift magnitudes. The sacrificial-band mechanism provides no measurable spectral gain for steady-state tonal signals at $L=2$.
- The benefit diminishes at $|\text{shift}| > 4$ semitones, where phase quantization errors exceed the isolation capacity of the $L=2$ sacrificial band. The piano +6 st case shows a -1.02 dB LSD regression, the only regression observed in the study.
- TIR is unreliable for soft-attack signal classes (flute, sustained vocal) where no sharp onset exists within the ± 25 ms analysis window.
- A dedicated ablation experiment ($2\times$ sample rate PV without anti-alias LPF) is required to formally separate the multirate distortion-migration benefit from the increased time-frequency resolution at $L\cdot f_s$.
- MUSHRA perceptual listening test has not yet been conducted. All results are computational metrics only.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

This project presented PAIRS (Phase Artifact Reduction Using Interpolation And Re-Sampling), a multirate preprocessing framework for reducing phase-vocoder-induced spectral distortion in pitch shifting. The core mechanism upsampling prior to PV processing to create a sacrificial high-frequency band into which intermodulation distortion migrates, followed by mandatory anti-aliasing LPF scrubbing requires no modification to the PV and is fully composable with any internal phase vocoder improvement.

The primary finding is that PAIRS consistently preserves transient integrity better than Std PV on harmonic and percussive signal classes, achieving TIR > 1.0 in 11 of 17 real recordings, with the strongest benefit on piano (TIR up to 1.47) and plucked-string signals (TIR up to 1.11). LSD evaluation on 18 synthesized signals confirms the multirate mechanism via LPF-only ablation, with improvement in 4 of 6 signal classes and zero regressions at ± 3 to ± 4 semitone practical range. At $L=2$, PAIRS adds only $\sim 8\%$ computational overhead, versus $\sim 485\%$ for Zero-Padded PV.

The results collectively demonstrate that PAIRS occupies a well-defined and favorable position in the trade-off space between transient integrity, phase coherence, spectral fidelity, and computational overhead. PAIRS is confirmed as a safe, drop-in wrapper for any standard phase vocoder implementation within the practical pitch-shifting range of ± 3 to ± 4 semitones.

6.2 Advantages of the System

- Drop-in compatibility: PAIRS wraps any unmodified phase vocoder and requires no knowledge of or access to PV internals.
- Low computational overhead: $\sim 8\%$ additional cost relative to PV FFT cost (lower bound), versus $\sim 485\%$ for ZP-PV.
- Composability: PAIRS is orthogonal to internal PV corrections such as phase locking or transient-aware routing.
- Zero regressions in the practical range: No LSD regressions at ± 3 to ± 4 semitones across all five instrument classes.
- Transient preservation: Consistent TIR > 1.0 on piano and plucked-string instruments, with highest observed TIR of 1.47 on piano at -3 semitones.
- Interpretable mechanism: The sacrificial-band argument provides a clear, testable mechanistic explanation grounded in standard multirate signal processing theory.

6.3 Future Scope

- MUSHRA (ITU-R BS.1534-3) perceptual listening test with minimum 15 experienced listeners as the primary prerequisite for journal submission.
- Formal statistical analysis of TIR and Δcoh across a full 18-recording RWC corpus with Holm-Bonferroni correction across all method pairs.
- Transient-aware PAIRS with onset-detection bypass during percussive attacks, to address transient smearing at large shift magnitudes.
- Upsampling factor sweep ($L = 2, 3, 4$) to characterize the overhead-vs-LSD trade-off and determine the optimal L for various use cases.
- Dedicated ablation experiment: $2\times$ sample rate PV without anti-alias LPF, as the highest-priority missing mechanistic validation.
- Real-time hardware implementation on embedded DSP platforms (ARM Cortex-M, TMS320C6748) for music production and prosthetic voice applications.
- Extension to polyphonic and multi-channel audio, including stereo pitch shifting with phase coherence across left and right channels.

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APPENDIX A

GLOSSARY OF TERMS

Table A.1: Glossary of Terms

Term	Full Form / Definition
PV / Phase Vocoder	A signal processing technique using STFT-based time-frequency analysis for pitch shifting and time stretching.
STFT	Short-Time Fourier Transform computes frequency spectrum over sliding time windows.
IMD	Intermodulation Distortion spurious frequency components generated when two or more frequencies interact nonlinearly.
Sacrificial Band	The high-frequency spectral region $[f_s/2, L \cdot f_s/2]$ left empty by the interpolation filter; receives migrated IMD during PV processing.
TIR	Transient Integrity Ratio ratio of Std PV transient error to PAIRS transient error at ± 25 ms around signal peak. $TIR > 1.0$ = PAIRS wins.
Δcoh	Inter-frame phase coherence gain difference in mean phase coherence between PAIRS and Std PV outputs.
LSD	Log Spectral Distance frame-based measure of spectral fidelity to an ideal reference. Lower = better.
Noble Identity	A theorem stating that LTI filtering and downsampling commute under specific conditions; does NOT apply to the nonlinear PV.
Polyphase FIR	An efficient filter structure for multirate signal processing that splits a FIR filter into sub-filters for each phase.
Kaiser Window	A parameterizable FIR design window; $\beta=8$ provides excellent stopband attenuation (~ 80 dB) used in PAIRS interpolation filter.
MUSHRA	MULTiple Stimuli with Hidden Reference and Anchor (ITU-R BS.1534-3); the standard perceptual listening test protocol for audio quality evaluation.
RWC Database	Real World Computing Music Database a standard benchmark dataset of musical instrument recordings used for audio DSP evaluation.

Table A.1: Glossary of Terms

APPENDIX B

EVALUATION METRIC FORMULAE

B.1 Log Spectral Distance (LSD)

LSD is computed as the mean squared log-spectral ratio between the reference and estimated power spectra:

$$\text{LSD} = (1/K) \cdot \sum_k [10 \cdot \log_{10}(\text{Sref}[k] / \text{Sest}[k])]^2$$

where $\text{Sref}[k]$ and $\text{Sest}[k]$ are the reference and estimated power spectral densities at bin k , and K is the number of frequency bins. The reference is constructed via MATLAB's rational resampler (polyphase FIR, sufficient order). Computation uses NFFT=1024, 50% overlap, Hann window.

B.2 Transient Integrity Ratio (TIR)

$$\text{TIR} = \text{RMS_err}(\text{Std PV}) / \text{RMS_err}(\text{PAIRS}) \quad [\text{at } \pm 25 \text{ ms transient window}]$$

where RMS_err is the root-mean-square error of the processed output relative to the original signal within a ± 25 ms window centred on the detected signal peak (max amplitude sample). $\text{TIR} > 1.0$ indicates PAIRS preserves the transient better than Std PV; $\text{TIR} < 1.0$ indicates Std PV performs better.

B.3 Inter-frame Phase Coherence Gain (Δcoh)

$$\Delta\text{coh} = \text{mean_coh}(\text{PAIRS}) - \text{mean_coh}(\text{Std PV})$$

where coherence per frame is computed as $|E[\exp(j \cdot \Delta\phi)]|$ across frequency bins, and $\Delta\phi$ is the inter-frame phase difference. $\Delta\text{coh} > 0$ indicates more stable inter-frame phase progression in the PAIRS output.